

Incremental LEFM Prediction of a Crack Trajectory from a Circular Hole

Pilot manuscript

Abstract

The incremental direction of a hole-initiated crack is examined in an asymmetric elastic plate using linear elastic fracture mechanics and the maximum tangential stress criterion. A stress-selected point on the circular hole defines the initiation location and a frozen material-side normal. Only the first crack segment is prescribed along that normal. Subsequent directions follow from the current-tip stress-intensity factors after one elastic solution of each fixed, complete crack geometry. A constant 4-mm increment produces an accepted history from 4 to 92 mm. The opening-mode intensity increases throughout this history, whereas positive mode mixity reaches a maximum at 68 mm and changes sign between the solved 84- and 88-mm states. This change reverses the sign of the incremental turn while the absolute crack direction remains negatively inclined. Reflection-paired tip meshes, prescribed-field controls, strict residual checks, and crack-opening-displacement comparisons support the numerical interpretation. A second trajectory, propagated independently with a substantially coarser exterior mesh while retaining the same paired tip core and interaction-domain region, preserves the P_{17} mode-mixity maximum and the P_{21} – P_{22} sign-change bracket; the maximum separation of corresponding accepted tips is only $0.0674\ \mu\text{m}$. The late redirection is consistent with increasing free-boundary influence rather than unloading, although these calculations do not isolate boundary effects from accumulated curvature. The next geometry is qualified but physically unsolved. The findings establish a verified fixed-increment pilot sequence, not increment-size convergence or a prediction of arrest, branching, or boundary intersection.

Keywords: crack trajectory; circular hole; linear elastic fracture mechanics; maximum tangential stress; mode mixity; interaction integral; crack-opening displacement.

1 Introduction

Stress concentrations around holes identify possible crack-initiation sites, but an initiation site alone does not determine the subsequent trajectory. Once a finite crack exists, its direction depends on the mixed-mode field at its current tip and on the entire existing crack geometry. A direction-selection calculation must therefore update the elastic problem as the crack advances, rather than assign the complete trajectory at initiation.

The present pilot considers a single crack growing from a circular hole in a finite, asymmetric plate. The scientific question is how its direction evolves when each new segment is determined only by the current-tip opening- and sliding-mode stress-intensity factors. The maximum tangential stress (MTS) rule supplies a local turn; accumulating that turn updates the absolute direction in a fixed initiation frame. The imposed finite increment specifies the geometric advance, while the stress-intensity factors determine its direction.

The central result is a change in the sign of mode II while mode I continues to increase. Positive mode mixity first grows, reaches a maximum, and then decreases through zero. The resulting change in turning sense occurs as the tip approaches the right free boundary. In addition to the reference calculation, an independently propagated trajectory with a substantially coarser

exterior mesh is used to test whether this sequence is sensitive to the exterior discretization. The comparison preserves the characteristic states and produces only submicrometer differences between corresponding accepted tips. It does not, however, establish increment-size convergence, local tip-mesh convergence, or a causal boundary-effect decomposition. Numerical verification is used to separate the physical interpretation from extraction artifacts and from the eventual linear-solver limit.

2 Problem formulation

2.1 Geometry, material, and loading

The plate occupies $[0, A] \times [-B, B]$, with $A = 300$ mm and $B = 100$ mm. A circular hole of radius $R = 30$ mm is centered at $(170, -20)$ mm. The finite-element hole boundary is represented by a 480-segment polygon with the fitted initiation point retained. The material is isotropic, with $E = 210000$ MPa and $\nu = 0.30$, under plane strain. For the crack geometry Γ_k ending at P_k , the elastic problem is

$$\nabla \cdot \boldsymbol{\sigma} = \mathbf{0} \quad \text{in } \Omega_k, \quad (1)$$

$$\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T), \quad \varepsilon_{33} = 0, \quad (2)$$

$$\boldsymbol{\sigma} = 2\mu\boldsymbol{\varepsilon} + \lambda \operatorname{tr}(\boldsymbol{\varepsilon})\mathbf{I}, \quad \mu = \frac{E}{2(1+\nu)}, \quad \lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}. \quad (3)$$

The top and bottom tractions are $(0, \sigma_0)$ and $(0, -\sigma_0)$, respectively, with $\sigma_0 = 1$ MPa. The vertical sides, hole boundary, and crack faces are traction free. Rigid-body motions are removed by imposing $u_x = u_y = 0$ at the lower left corner and $u_y = 0$ at the lower right corner. There are no body forces.

All reported SIFs correspond to this unit traction and have units $\text{MPa}\sqrt{\text{m}}$. For a fixed geometry, positive scaling of the traction scales both SIFs equally and preserves their ratio and the MTS direction. The sequence studied here is a direction-selection calculation under normalized loading, not a load-versus-time fracture-growth history.

2.2 Stress-selected initiation and the frozen frame

The uncracked-hole elastic field selects the initiation point from the maximum positive tangential boundary stress. In the recorded initiation procedure, T6 stresses sampled at three material-side offsets are extrapolated linearly to the boundary, followed by a local quadratic angular fit. With tensile threshold $\sigma_c = 300$ MPa, the normalized initiation rule is

$$\phi_* = \arg \max_{\phi} \sigma_{tt}^+(\phi; \sigma_0), \quad \lambda_{\text{ini}} = \frac{\sigma_c}{\max_{\phi} \sigma_{tt}^+(\phi; \sigma_0)}, \quad \sigma_{tt}^+ = \max(\sigma_{tt}, 0). \quad (4)$$

The accepted fitted angle is $\phi_* = -1.5606127781^\circ$. The initiation point is

$$P_0 = (0.199988872196, -0.0208170339049) \text{ m}. \quad (5)$$

The fitted peak stress is 3.714071 MPa at unit loading, giving $\lambda_{\text{ini}} = 80.773906$. This initiation condition supplies the location; it is not reused as a growth or arrest criterion along the crack.

The frozen basis is

$$\mathbf{n}_{\text{mat}} = \begin{bmatrix} 0.999629073206 \\ -0.0272344634968 \end{bmatrix}, \quad \mathbf{t} = \begin{bmatrix} -n_{\text{mat},y} \\ n_{\text{mat},x} \end{bmatrix}. \quad (6)$$

The normal points radially out of the hole and into the material; $\mathbf{t}$ is its counterclockwise rotation. All absolute crack angles θ_k below are measured relative to this frozen normal, not to the global

x axis. The first segment is prescribed by

$$\Delta a = 4 \text{ mm}, \quad \theta_1 = 0, \quad P_1 = P_0 + \Delta a \mathbf{n}_{\text{mat}}. \quad (7)$$

An earlier first-angle sweep was an independent verification exercise and does not determine θ_1 in this formulation. The study retains one active crack; subsequent competing initiation sites are not modeled.

3 Incremental crack-path formulation

3.1 Current-tip frame and stress-intensity factors

At P_k , the local forward direction is the *last* segment direction, $\mathbf{e}_1^{(k)} = (P_k - P_{k-1})/\Delta a$, with $\mathbf{e}_2^{(k)} = (-e_{1y}^{(k)}, e_{1x}^{(k)})$. This frame is not the mouth-to-tip chord. With r measured ahead of the tip along x_1 , the signed conventions correspond to

$$K_I^{(k)} = \lim_{r \rightarrow 0^+} \sqrt{2\pi r} \sigma_{22}(r, 0), \quad K_{II}^{(k)} = \lim_{r \rightarrow 0^+} \sqrt{2\pi r} \sigma_{12}(r, 0). \quad (8)$$

These are evaluated numerically by the interaction integral described below. The complete previous crack geometry is retained during each elastic solution.

3.2 The implemented MTS rule

Let α be a turn relative to $\mathbf{e}_1^{(k)}$, and write $c = \cos(\alpha/2)$ and $s = \sin(\alpha/2)$. The implemented angular coefficients are exactly

$$S(\alpha) = K_I^{(k)} c^3 - 3K_{II}^{(k)} s c^2, \quad (9)$$

$$T(\alpha) = K_I^{(k)} s c^2 + K_{II}^{(k)} c(1 - 3s^2). \quad (10)$$

Their physical stress amplitudes carry the common factor $1/\sqrt{2\pi r}$. Since $S' = -3T/2$, the direction rule finds a tensile stationary direction by solving $T(\alpha) = 0$. In the implementation the root lies in $[-80^\circ, 80^\circ]$: a bracket of $\pm 10^\circ$ around a clipped initial estimate $-2K_{II}/K_I$ is used first and expanded to the full interval if necessary. For $|K_I| < 10^{-12}$ the initial estimate is $\text{sign}(K_{II})70.5^\circ$ instead. The routine sets $\alpha = 0$ when $|K_{II}| < 10^{-12}$ in the stored SIF units, and includes an opposite-angle safeguard if the computed circumferential coefficient is compressive. The accepted history has $K_I > 0$ and small tensile-branch turns. No angular scan is performed during propagation.

For small mode mixity, expansion of Eq. (10) gives $\alpha \simeq -2K_{II}/K_I$ in radians. This relation explains the turning sign but does not replace the root calculation used for the results.

3.3 Finite-increment recurrence

Define the direction vector in the frozen frame by

$$\mathbf{e}(\theta) = \cos \theta \mathbf{n}_{\text{mat}} + \sin \theta \mathbf{t}. \quad (11)$$

The incremental and absolute angles are distinguished in the recurrence

$$\Delta \theta_{k+1} = \alpha_{\text{MTS}}(K_I^{(k)}, K_{II}^{(k)}), \quad (12)$$

$$\theta_{k+1} = \theta_k + \Delta \theta_{k+1}, \quad (13)$$

$$P_{k+1} = P_k + \Delta a \mathbf{e}(\theta_{k+1}). \quad (14)$$

Thus each fixed current geometry is qualified, solved once, and used to append one segment. Only the first direction and the increment length are prescribed. The accepted first-tip SIFs are $K_I^{(1)} = 0.366479612185$ and $K_{II}^{(1)} = 4.19826648316 \times 10^{-6}$, which give $\Delta\theta_2 = \theta_2 = -0.001312722142^\circ$. This previously qualified first-tip state seeds the long run; the accepted subsequent elastic states are P_2 through P_{23} .

4 Numerical implementation and verification

4.1 Geometry and paired tip discretization

The crack is represented by its complete polyline with distinct upper and lower face topology, including previous kink vertices. A temporary appended-hole carrier supports geometric identification; its crack faces are collapsed to the zero-width polyline while preserving separate face identities. A structured core is aligned with the last segment and paired under local reflection across that segment. The asymmetric physical exterior is retained through conforming graded triangulation. T3 triangles are converted to straight-sided quadratic T6 elements.

The fixed production scales are

$$\frac{h_{\text{tip}}}{\Delta a} = 0.00675308135, \quad \frac{r_i}{\Delta a} = 0.10, \quad \frac{r_o}{\Delta a} = 0.65, \quad \frac{r_c}{\Delta a} = 0.75. \quad (15)$$

For this 4-mm increment these give $h_{\text{tip}} = 0.0270123$ mm, $r_i = 0.4$ mm, $r_o = 2.6$ mm, and $r_c = 3$ mm. The grading transition is 4 mm and the far-field size cap is 2.5 mm. The paired core contains 12678 T3 triangles. All primary FE-nodal integration support lies inside this core, with a fixed fingerprint of 11316 participating elements.

Geometry and mesh checks retain the physical boundary vertices, distinct crack faces, shared intact ligament, conforming seams, positive element areas and T6 Jacobians. The mesh-angle and adjacent longest-edge-ratio gates are 20° and 1.8, respectively. These checks establish discretization integrity, not mesh-independent physical predictions.

For an exterior-mesh sensitivity test, a second mesh family, denoted M1, retains the same paired tip core, h_{tip} , r_i , r_o , and r_c , together with the same 4-mm transition length and all qualification gates. Only the exterior size law is relaxed: the far-field size cap is increased from 2.5 to 5.0 mm, the far-field grading slope from 0.10 to 0.15, and the boundary-metric growth parameter from 0.25 to 0.35. On the fixed P_{17} and P_{22} geometries this reduces the total T3 count by about 31–33% and the exterior-element count by about 40–43%, while leaving the 12678-element paired core and the 11316-element interaction-domain support unchanged. These fixed-geometry checks were used only to qualify M1 for a subsequent independent trajectory calculation.

4.2 Interaction/domain integral

For actual field $(\boldsymbol{\sigma}, \boldsymbol{\varepsilon}, \mathbf{u})$ and a unit auxiliary mode m , the implemented local-frame density gives

$$I^{(m)} = \int_{\Omega_k} \left[-(\boldsymbol{\sigma}^{(m)} : \boldsymbol{\varepsilon}) \delta_{1j} + \sigma_{ij} u_{i,1}^{(m)} + \sigma_{ij}^{(m)} u_{i,1} \right] q_{,j} dA, \quad i, j \in \{1, 2\}. \quad (16)$$

Actual gradients use the T6 displacement field. Analytical near-tip auxiliary stresses and centered finite-difference derivatives of auxiliary displacements are used, with the recorded derivative step $h = \max(10^{-7} \text{ m}, 10^{-5} r)$. A radial nodal weight is one for $r \leq r_i$, decreases linearly to zero at r_o , and is zero outside. Its T6 interpolant supplies the FE-consistent gradient in Eq. (16). Integration uses the 16-point triangular rule. For auxiliary amplitude $K_{\text{aux}} = 1$ and plane strain,

$$K_m = \frac{E'}{2K_{\text{aux}}} I^{(m)}, \quad E' = \frac{E}{1 - \nu^2}, \quad m \in \{I, II\}. \quad (17)$$

The normalization and sign convention are those used in the recorded extraction procedure.

4.3 Independent COD postprocessing

Native upper/lower T6 face nodes are paired in the last-segment frame. If $\llbracket u_i \rrbracket$ denotes the upper-minus-lower displacement jump behind the tip at distance r , the apparent COD SIFs are

$$\begin{bmatrix} K_I^{\text{app}}(r) \\ K_{II}^{\text{app}}(r) \end{bmatrix} = \frac{\mu}{\kappa + 1} \sqrt{\frac{2\pi}{r}} \begin{bmatrix} \llbracket u_2 \rrbracket \\ \llbracket u_1 \rrbracket \end{bmatrix}, \quad \kappa = 3 - 4\nu. \quad (18)$$

Linear and quadratic fits in $r/\Delta a$ extrapolate these quantities to $r = 0$. The four fitting windows are $[0.04, 0.20]$, $[0.04, 0.30]$, $[0.08, 0.30]$, and $[0.12, 0.30]$ times Δa , with 38, 55, 44, and 34 native samples. COD verifies a second extraction route from the *same* elastic field. The production direction always uses the EDI SIFs; window/order scatter is not a statistical uncertainty interval.

4.4 Solver and evidence checks

The unclamped symmetric stiffness is reduced to free degrees of freedom, symmetrized within the audited formulation, and reordered by a symmetric approximate minimum-degree permutation. Parameter-free symmetric Gauss–Seidel preconditioning is used with conjugate gradients. The reported-residual tolerance is 10^{-10} , the iteration limit is 5000, and the independently computed true-relative-residual gate is 5×10^{-10} . Constrained degrees of freedom remain exactly zero. Only an accepted elastic field is checkpointed, before SIF/COD/MTS postprocessing.

Prescribed nodal pure-I, pure-II, and tiny-mixed $(K_I, K_{II}) = (1, 10^{-4})$ fields test modal recovery and leakage on each qualified mesh. Recovery and tiny-mixed relative-error gates are 2×10^{-4} ; the pure-mode cross-leakage and superposition gates are 10^{-10} . The observed maximum tiny-mixed relative error is 3.86×10^{-8} . These controls qualify extraction capability and topology rather than the global physical finite-element error.

A short regression reproduces a previously accepted two-segment P_2 state before extending the path. Historical straight 8-mm calculations provided separate extraction and controlled-mesh verification; they are not points on the propagated trajectory and do not establish convergence of the present 4-mm path.

The authoritative long-run archive contains 22 accepted compact physical records (P_2 – P_{23}) and 23 qualification records (P_2 – P_{24}). A semantic audit against the atomic path state confirms the accepted SIFs, mode mixity, turns, absolute directions, and geometry. The maximum direction-record difference is 4.36×10^{-14} degrees; the other compared scientific quantities agree exactly. The maximum accepted true residual is 9.9963×10^{-11} , and iteration counts range from 2442 to 3011. Supporting diagnostics are given in Fig. S1.

5 Crack-trajectory results

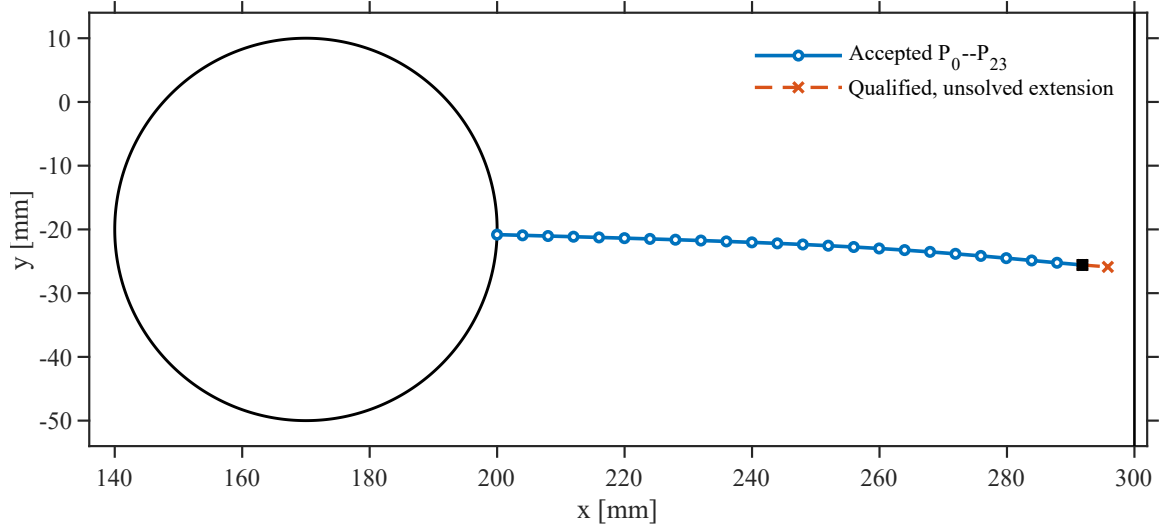
5.1 Accepted path and direction history

Figure 1 distinguishes the accepted P_0 – P_{23} polyline from the dashed, qualified but unsolved P_{23} – P_{24} extension. The accepted crack length is 92 mm. The last tip and its horizontal clearance from the right boundary are

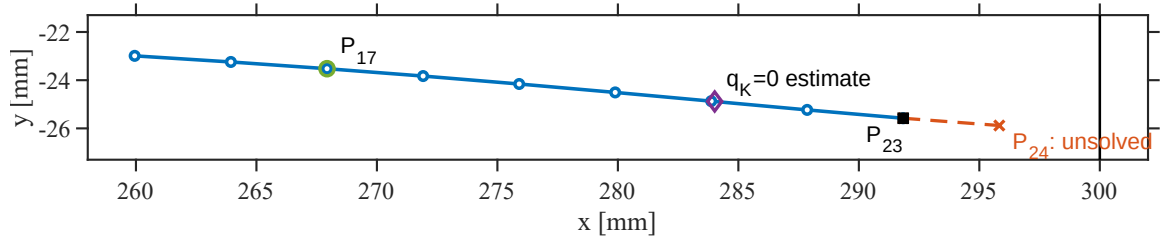
$$P_{23} = (291.840484, -25.578941) \text{ mm}, \quad A - x_{23} = 8.1595 \text{ mm}. \quad (19)$$

Directions vary gradually, although the finite-increment representation is piecewise linear rather than a continuously smooth curve.

The absolute direction becomes increasingly negative up to $\theta_{22} = -3.645964^\circ$, then rotates toward a less negative inclination at P_{23} (Fig. 2). A positive late turn does not imply an upward-moving tip: the accepted segments remain negatively inclined and their global y coordinates continue to decrease.



(a) Hole and accepted trajectory through P_{23} , with the qualified but physically unsolved P_{23} – P_{24} extension.



(b) Late-path detail with equal spatial scales. The markers identify P_{17} , the linearly interpolated $K_{II}/K_I = 0$ location, P_{23} , and the qualified unsolved point P_{24} .

Figure 1: Hole-initiated trajectory and late detail. Both spatial panels use equal x and y scales within the panel. The circle gives geometric context; the numerical hole is polygonally discretized. Solid segments are accepted geometry through P_{23} . The dashed extension is qualified geometry only and has no accepted physical SIFs. The diamond is a linear interpolation marker, not a solved tip.

Table 1: Characteristic accepted states. SIFs have units $\text{MPa}\sqrt{\text{m}}$; angles are in degrees. The interpolated row contains only a graphical zero estimate, not a solved stress field or crack direction.

State	a [mm]	K_I	K_{II}	K_{II}/K_I	θ_k	$\Delta\theta_{k+1}$
P_1	4	0.366480	4.1983×10^{-6}	1.1456×10^{-5}	0.000000	-0.001313
P_{17}	68	0.786793	2.5484×10^{-3}	3.2389×10^{-3}	-2.444435	-0.371145
P_{21}	84	0.966310	7.8602×10^{-5}	8.1343×10^{-5}	-3.636643	-0.009321
P_{22}	88	1.066724	-2.2845×10^{-3}	-2.1416×10^{-3}	-3.645964	+0.245406
P_{23}	92	1.244360	-6.7803×10^{-3}	-5.4488×10^{-3}	-3.400558	+0.624347
Interpolated	84.15	—	—	0	—	—

5.2 Mode intensities and the positive maximum

Figure 2 shows monotonically increasing K_I , from 0.366480 at P_1 to 1.244360 at P_{23} . The initially very small positive K_{II} grows to its positive maximum at P_{17} , where $a = 68$ mm, $K_{II} = 0.002548369$, and $q_K = K_{II}/K_I = 0.003238931$. The positive mode-mixity maximum occurs at the same state. It is not the maximum *absolute* mode-II intensity: the negative mode-II magnitude at the last accepted tip is larger.

From P_{17} through P_{21} , K_{II} remains positive but decreases. Accordingly the negative incremental MTS turn becomes smaller in magnitude (Fig. 3). This change in turning response accompanies an increasing opening-mode intensity, not unloading.

5.3 Local-symmetry crossing and late turning reversal

At P_{21} ($a = 84$ mm), $q_K = +8.134267 \times 10^{-5}$; at P_{22} ($a = 88$ mm), $q_K = -0.002141592$. The sign change is therefore bracketed by two solved states. For visualization only, linear interpolation gives

$$a_{\text{LS}} = a_{21} - \frac{q_K^{(21)}}{q_K^{(22)} - q_K^{(21)}}(a_{22} - a_{21}) \simeq 84.15 \text{ mm}. \quad (20)$$

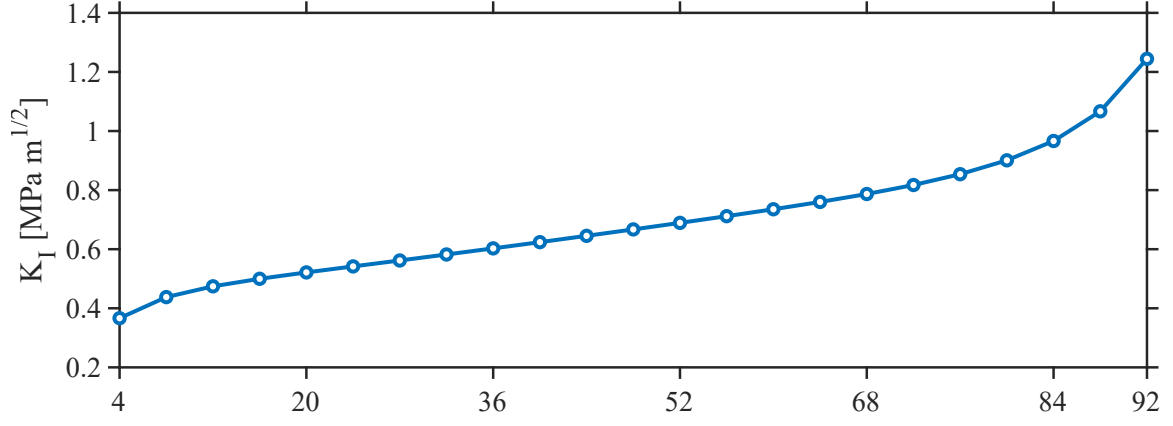
This estimate is not an independently solved local-symmetry state and does not quantify interpolation or increment-size error.

Mode II is negative at P_{22} and P_{23} , so the next turning increments are positive. The last accepted state has $K_I = 1.244360444$, $K_{II} = -0.006780312$, and $q_K = -0.005448833$. It predicts $\Delta\theta_{24} = +0.624347038^\circ$ and $\theta_{24} = -2.776211097^\circ$. The latter defines the next qualified geometry, not a physical solution at P_{24} .

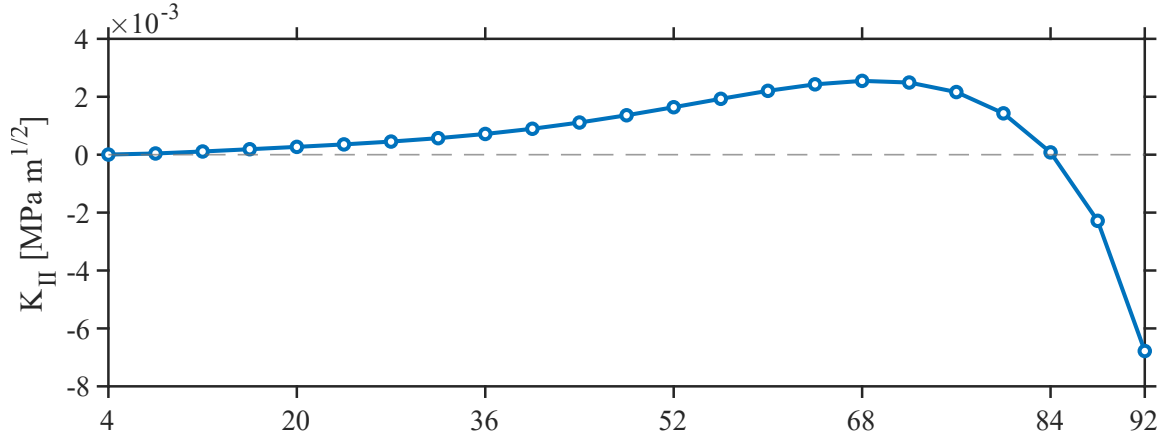
5.4 COD comparison and extraction sensitivity

All eight stored COD fit definitions retain positive mode mixity at P_{21} and negative mode mixity at P_{22} , corroborating the bracketed sign change without making COD the direction selector. Differences remain nonzero and depend on the fitting definition (Fig. 4). Over all 176 fits, the largest absolute COD–EDI turning difference is 0.010236° , at P_{23} for a linear fit over $[0.12, 0.30]\Delta a$. The quadratic-fit turns at P_{23} range from 0.620436° to 0.622478° , compared with the EDI value 0.624347038° .

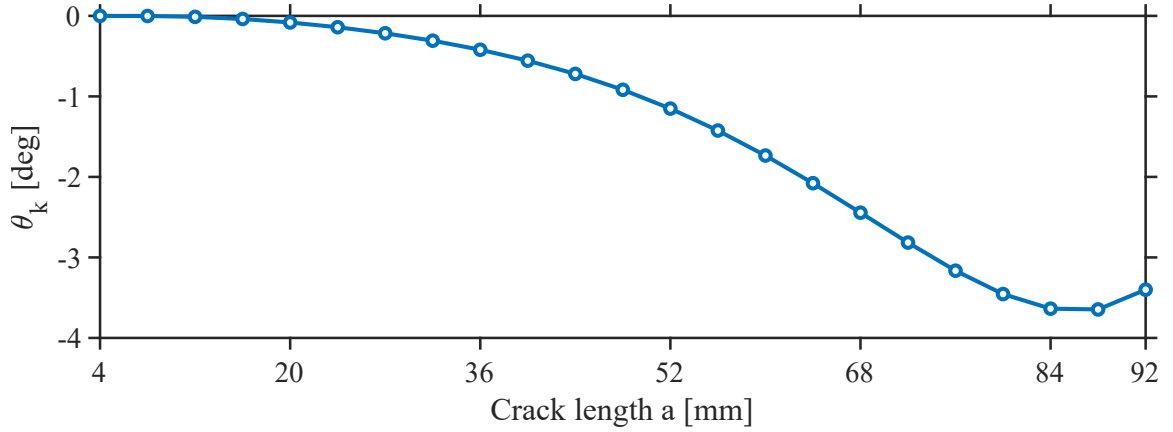
Near local symmetry, relative differences can be large despite small absolute errors. At P_{21} the same rear-window linear fit gives mode mixity about 26.3% above EDI, while its turning-angle difference is only 0.002453° . Thus visual overlap of full-history curves is not sufficient to establish extraction accuracy. The late fitting sensitivity is a postprocessing limitation to investigate, not an error bar on the physical path. Both routes also share any error in the underlying elastic solution.



(a) Opening-mode stress-intensity factor K_I .

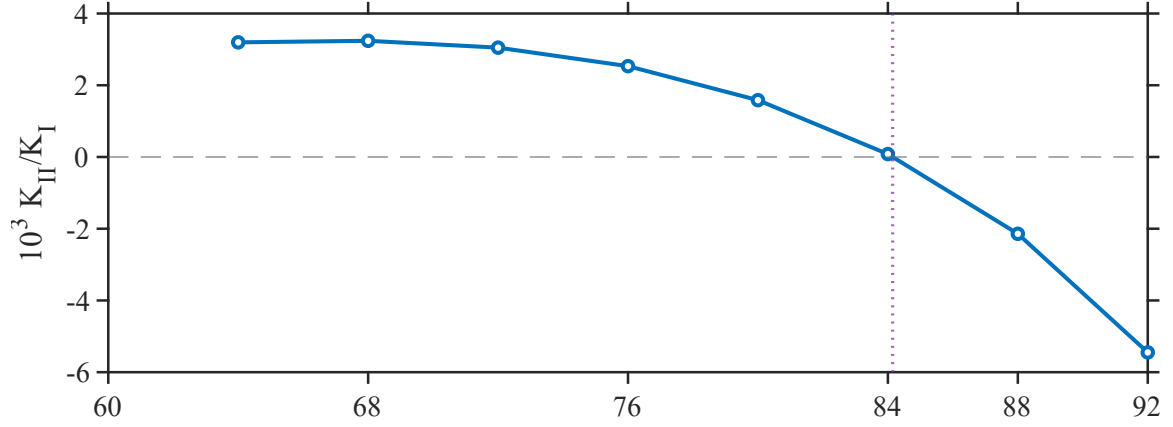


(b) Sliding-mode stress-intensity factor K_{II} .

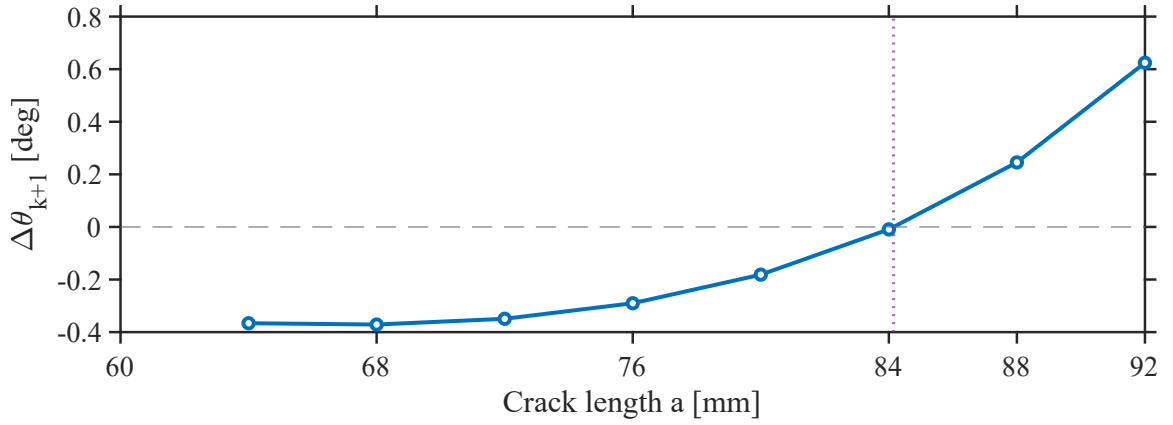


(c) Absolute crack direction θ_k in the frozen initiation frame.

Figure 2: Accepted intensities and absolute direction. Intensities correspond to unit remote traction. The direction minimum occurs at P_{22} ; positive turning there produces the less negative P_{23} direction. No physical curve includes P_{24} .

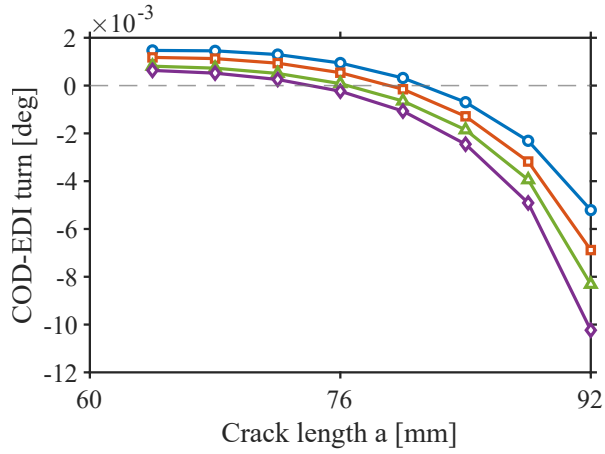


(a) Late mode-mixity history.

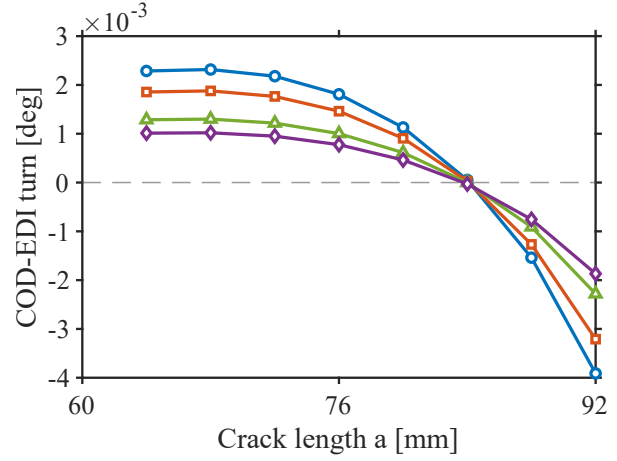


(b) Incremental MTS turn relative to the current segment.

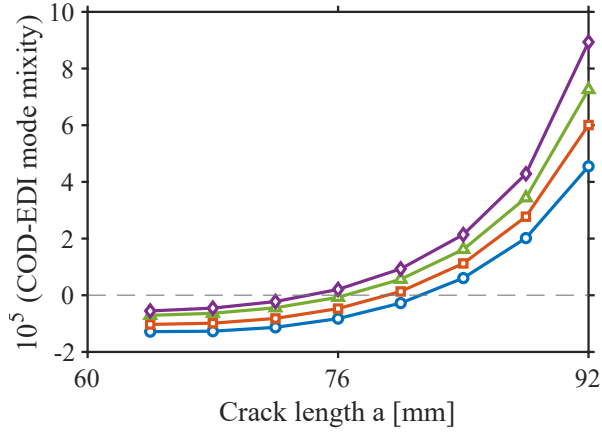
Figure 3: Mode-mixity sign reversal and opposite-signed MTS turning. Each marker is an accepted current-tip state; its turn predicts the next segment. The vertical dotted line is the interpolated $K_{II}/K_I = 0$ location. The positive turn reported at $a = 92$ mm belongs to the P_{23} solution and predicts the unsolved next geometry.



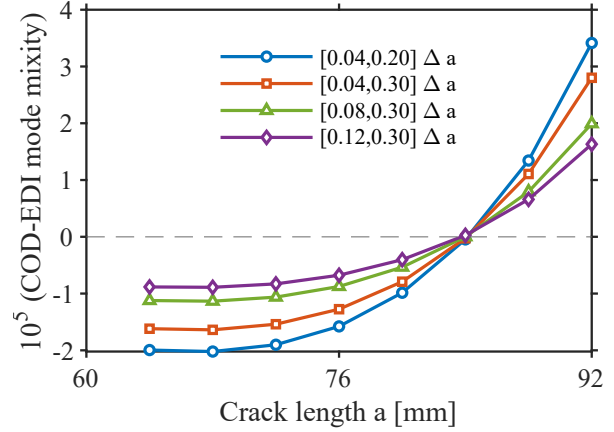
(a) Turning-angle differences for linear COD fits.



(b) Turning-angle differences for quadratic COD fits.



(c) Mode-mixity differences for linear COD fits.



(d) Mode-mixity differences for quadratic COD fits.

Figure 4: Late-path extraction differences for all four native COD windows and both polynomial degrees. Window bounds are multiples of Δa . Zero is equality with EDI, not a known exact solution. The upper panels show differences in the MTS turns computed from the two sets of SIFs; the lower panels show differences in mode mixity. The largest observed difference occurs at the last accepted tip; individual difference curves need not vary monotonically.

5.5 Sensitivity to exterior-mesh density

The fixed-geometry M1 checks establish that the coarser exterior mesh does not materially perturb the extracted current-tip quantities when the crack polyline is held fixed. To test accumulated trajectory sensitivity, however, the M1 path was regenerated independently rather than evaluated on the reference vertices. The prescribed first segment P_0P_1 is common to both calculations, but the P_1 elastic problem is resolved on M1, its own $K_I^{(1)}$ and $K_{II}^{(1)}$ determine θ_2 , and every subsequent vertex follows recursively from the preceding M1 solution. Thus no reference-path vertex after P_1 is imposed.

Figure 5 compares the two complete accepted histories through P_{23} . At specimen scale the trajectories are visually indistinguishable [Fig. 5a]. The accumulated difference is resolved by plotting the coordinate and direction differences directly. Over all common accepted states, the maximum vertical-coordinate difference is $0.06734 \mu\text{m}$, the maximum Euclidean separation of corresponding tips is $0.06742 \mu\text{m}$, and the maximum absolute direction difference is 0.07304 millidegrees, or 7.30×10^{-5} degrees [Figs. 5b and 5c].

The mechanical history is equally stable. Both calculations place the positive maximum of K_{II}/K_I at P_{17} ($a = 68 \text{ mm}$), and both retain positive mode mixity at P_{21} and negative mode mixity at P_{22} [Fig. 5d]. The maximum absolute difference in K_{II}/K_I over the accepted histories is 2.89×10^{-7} . Linear interpolation of the two respective P_{21} – P_{22} brackets gives $a_{\text{LS}} = 84.146369912 \text{ mm}$ for the reference trajectory and 84.146048176 mm for M1, a difference of 0.000322 mm ($0.322 \mu\text{m}$). This sub-increment difference is reported only as a sensitivity diagnostic; neither interpolated value has physical precision at the micrometer scale.

The result therefore supports robustness of the reported mode-mixity maximum, sign reversal, and late turning response with respect to a substantial coarsening of the *exterior* mesh. It is deliberately narrower than a general mesh-independence claim: the paired tip core and interaction-domain radii are held fixed, so local tip-resolution and interaction-domain sensitivity remain separate questions.

6 Discussion

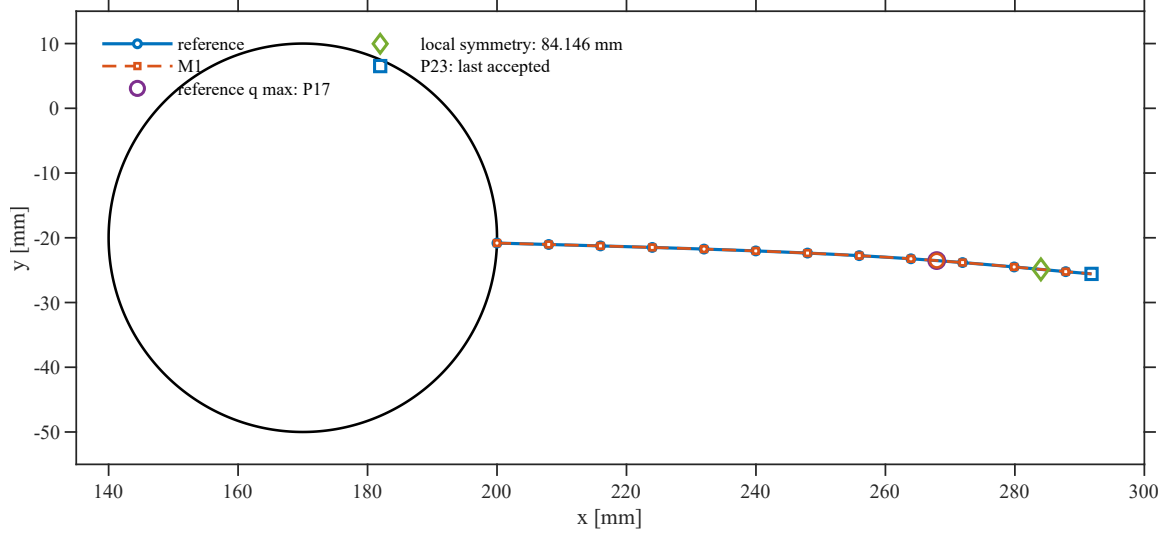
6.1 Interpretation of late redirection

The accepted sequence shows reorientation under increasing K_I . The change in turning sign is directly associated with the change in K_{II}/K_I , as required by Eqs. (10)–(13). The shrinking right-side ligament makes increased free-boundary influence a consistent physical interpretation. However, the boundary distance and the accumulated crack geometry change together. A control with a more distant boundary, or matched fixed-geometry calculations, is needed to isolate that influence. The result supports a mode-mixity explanation of the computed redirection; it does not prove that the boundary is its sole cause.

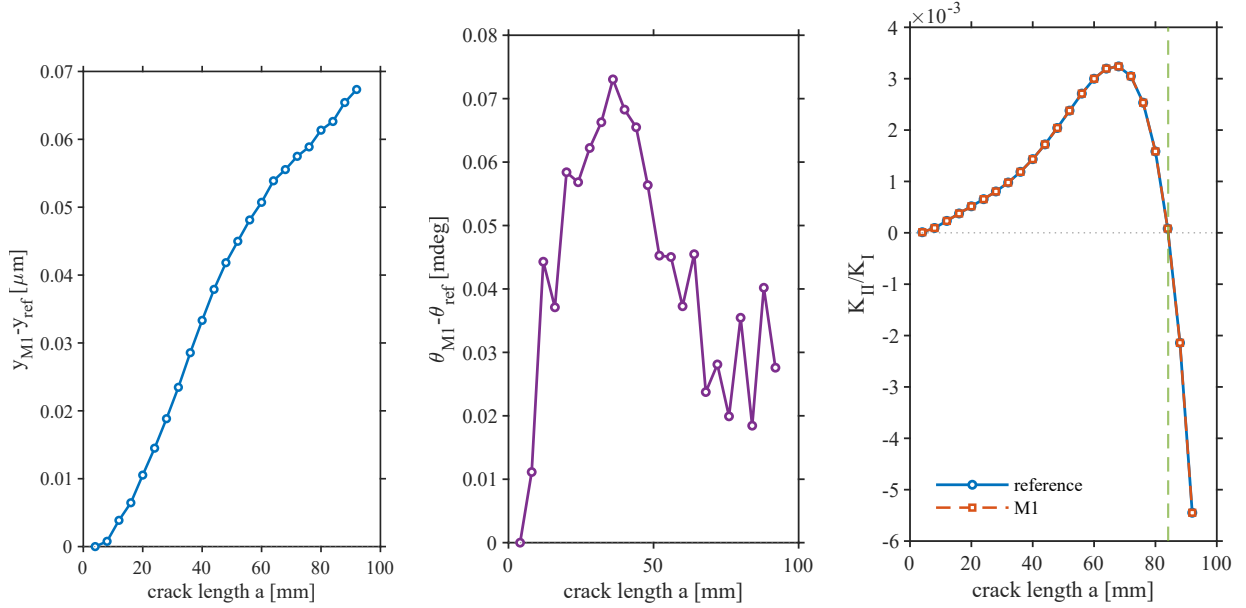
6.2 Discretization and boundary proximity

The directions are selected from the field before adding a finite straight segment. A 4-mm step therefore samples a continuously changing boundary-value problem at discrete geometries. No increment-size study is available for this path. The interpolated zero in Eq. (20) should be read against the 4-mm spacing of its bracketing states, not as a highly resolved physical crack length.

The independent M1 calculation addresses one part of the discretization question: substantial coarsening of the exterior mesh has negligible effect on the accepted path when the audited local tip construction is retained. It does not vary the tip-core resolution or the interaction-domain radii, and therefore cannot bound those contributions to path uncertainty. A smaller-increment family, together with controlled local tip-mesh and interaction-domain variations, is still required.



(a) Actual crack trajectories obtained with the reference exterior mesh and the independently propagated M1 exterior mesh. The two paths are visually indistinguishable at the scale of the specimen.



(b) Accumulated vertical tip-coordinate difference, $y_{M1} - y_{ref}$.

(c) Difference in the absolute local crack direction, $\theta_{M1} - \theta_{ref}$.

(d) Mode-mixity ratio K_{II}/K_I along the two independently propagated trajectories.

Figure 5: Sensitivity of the incremental LEFM trajectory to coarsening of the exterior mesh. The M1 calculation retains the audited crack-tip core and EDI extraction region but uses a substantially coarser exterior mesh. The comparison is based on two independently propagated trajectories: the M1 calculation recomputes the P_1 SIFs and every subsequent MTS update rather than reusing the reference vertices.

Local reflection pairing and synthetic recovery suppress known extraction leakage mechanisms, but do not by themselves bound the physical discretization error. Similarly, residual and element-quality gates measure numerical acceptance, not fracture-mechanics convergence. The historical 8-mm verification cannot be transferred into a general mesh-independence claim for this evolving polyline.

6.3 Last accepted state and model scope

The geometry ending at P_{24} , with total polyline length 96 mm, passes geometric and prescribed-field qualification. It has no accepted physical record. The reported attempted PCG solution did not satisfy the strict acceptance criterion; the available compact archive does not retain a failure trace that would quantify its stagnation. The physical history therefore ends at P_{23} , without retrospectively relaxing the gate. This is a numerical limitation, not evidence of fracture arrest or boundary intersection.

The model is quasi-static and linearly elastic. Crack advances are imposed for direction selection; no fracture-toughness growth law, dynamic instability, branching rule, plastic-zone assessment, or subsequent multicrack initiation calculation is provided. These restrictions define the scope of the present trajectory rather than mechanisms excluded by the results.

7 Conclusions

A stress-selected circular-hole point and one prescribed normal segment provide the initial state for an incremental trajectory whose later directions are determined by current-tip EDI SIFs and the implemented MTS rule. The accepted fixed-increment history reaches 92 mm at P_{23} . Positive mode mixity peaks at P_{17} , crosses zero between P_{21} and P_{22} , and becomes negative while opening-mode intensity continues to increase. This changes the sign of the incremental turn and reduces the negative absolute inclination late in the history.

The sign reversal is shared by all stored COD fitting definitions, although their late-path differences show finite extraction sensitivity. An independently propagated calculation with a substantially coarser exterior mesh reproduces the P_{17} mode-mixity maximum, the P_{21} – P_{22} sign-change bracket, and the accepted trajectory through P_{23} ; the maximum separation of corresponding tips is only 0.0674 μm . This supports robustness with respect to exterior-mesh density while leaving local tip-mesh and interaction-domain sensitivity unresolved. Paired-core, synthetic, solver, and archive checks establish a reproducible numerical sequence within the chosen discretization. The qualified unsolved extension must remain separate from the physical results. Increment-size convergence, controlled local tip/interaction-domain checks, and a boundary comparison are the next requirements for stronger predictive and causal claims.

References

Pilot status: verified literature entries have not yet been assembled. Explicit literature TODOs are retained in the source and in the accompanying bibliography file; no unverified citation is supplied.

A Supplementary numerical diagnostics

The accepted solver and mesh checks are summarized in Fig. S1. They provide reproducibility and acceptance diagnostics, rather than an estimate of the discretization error in the trajectory.

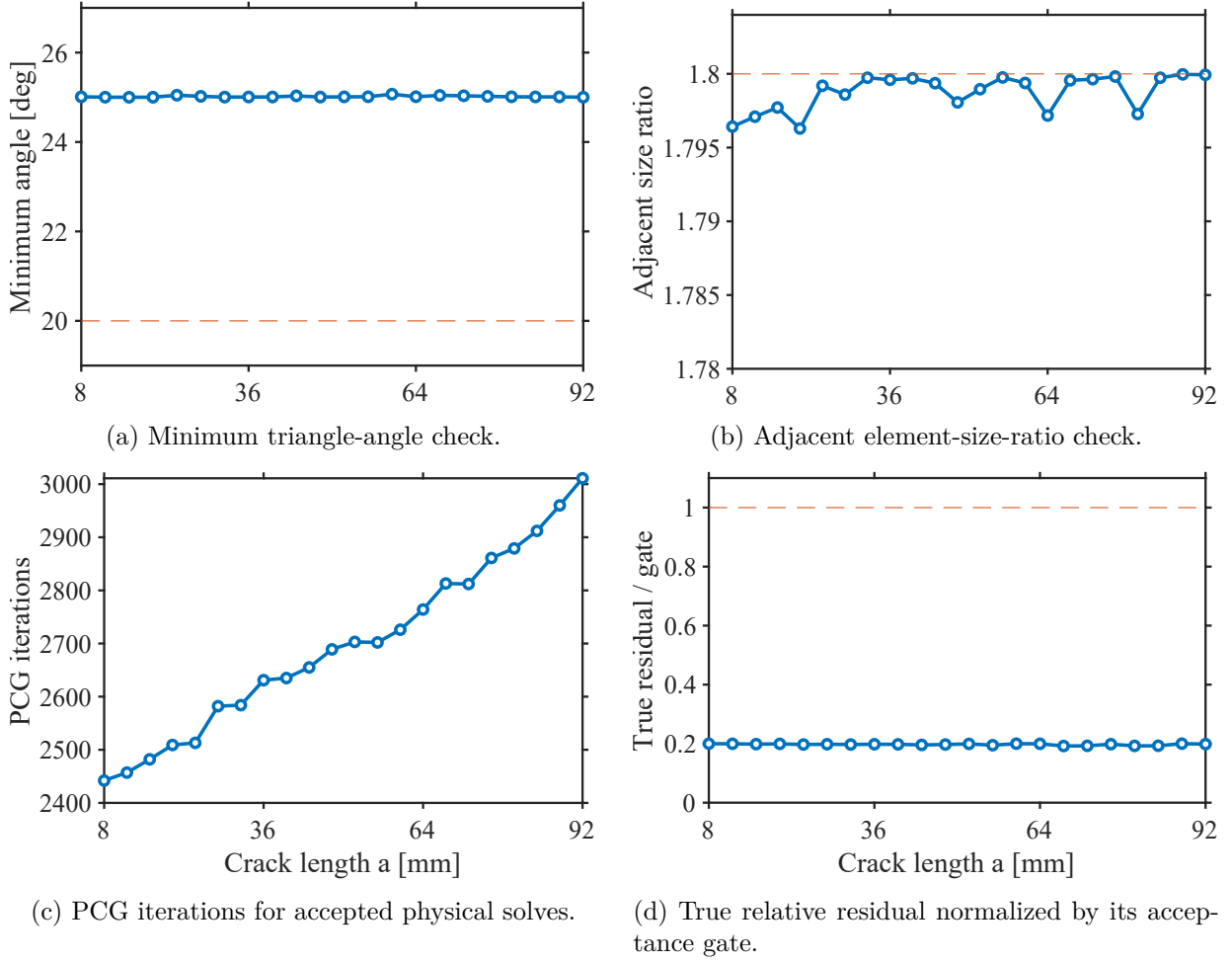


Figure S1: Diagnostics for accepted P_2 – P_{23} states. Dashed lines are the 20° minimum-angle gate, 1.8 adjacent-size-ratio gate, and normalized true-residual gate of one. The separately enforced reported-PCG tolerance is 10^{-10} ; the true-residual acceptance denominator is 5×10^{-10} . No failed P_{24} solve is plotted.